

3.2 DV Aufg.

$$b_1 = \frac{1}{\|1\|} = \frac{1}{\sqrt{\int_1^5 1 dx}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$\langle b_1, u_2 \rangle = \int_1^5 \frac{1}{2} x dx = \frac{1}{4} (24) = 6 \quad \rightarrow \frac{x-6}{\|x-6\|} = \frac{x-6}{\sqrt{\int_1^5 (x-6)^2}}$$

$$= \frac{x-6}{\sqrt{\int_1^5 x^2 - 12x + 36}} = \frac{x-6}{\sqrt{\left[\frac{x^3}{3} - 12 \cdot \frac{x^2}{2} + 36x\right]_1^5}} = \frac{x-6}{\sqrt{\frac{125}{3} - 150 + 180 - \frac{1}{3} + 6 - 36}}$$

$$= \frac{x-6}{\sqrt{\frac{124}{3}}} = \frac{x-6}{\sqrt{\frac{124}{3}}}$$

$$\rightarrow x - 6 \cdot \frac{1}{2} = x - 3 \quad \|x-3\| = \sqrt{\int_1^5 x^2 - 6x + 9} =$$

$$\sqrt{\left[\frac{x^3}{3} - \frac{6x^2}{2} + 9x\right]_1^5} = \sqrt{\left(\frac{125}{3} - 75 + 45 - \frac{1}{3} + 3 - 9\right)} = \frac{4}{\sqrt{3}}$$

$$b_2 = \frac{(x-3)\sqrt{3}}{4}$$

$$b_1 \langle b_1, u_3 \rangle + b_2 \langle b_2, u_3 \rangle = \frac{1}{2} \int_1^5 \frac{x^2}{2} dx + \frac{\sqrt{3}}{4} (x-3) \int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 dx$$

$$= 6(x-3) + \frac{31}{3}$$

$$u_3 - \left(6(x-3) + \frac{31}{3}\right) = x^2 - 6(x-3) - \frac{31}{3}$$

$$\|x^2 - 6(x-3) - \frac{31}{3}\| = 16$$

$$b_3 = \frac{x^2 - 6x + 18 - \frac{31}{3}}{\frac{16}{3\sqrt{5}}} = \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right)$$

$$B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right) \right)$$

$$\begin{aligned} \text{Proj}(\sin x) &= \frac{1}{2} \int_1^5 \frac{\sin x}{2} dx + \frac{\sqrt{3}}{4}(x-3) \int_1^5 \sin x \left(\frac{\sqrt{3}}{4}(x-3) \right) dx + \\ &+ \frac{3\sqrt{5}}{16} \int_1^5 \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right) \sin x dx = \\ &\approx (x^2 - 6x + \frac{23}{3}) \end{aligned}$$

$$\approx -0,0525(x^2 - 6x + 7,666) - 0,6466(x-3) + 0,06416 \approx$$

$$\approx \frac{1}{400} (-21x^2 - 132,68x + 640,585) \approx -0,0525x^2 - 0,33x + 1,601$$

$$\|\sin x + 0,0525x^2 + 0,33x - 1,601\| = \sqrt{\int_2^5 (\sin x + 0,0525x^2 + 0,33x - 1,601)^2 dx} \approx 0,32$$

$$=$$